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Mussel-inspired polymeric coatings with the
antifouling efficacy controlled by topologies†

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Block copolymers with different topologies (linear, loop, 3-armed and 4-armed polymers) containing poly(*N*-vinylpyrrolidone) (PVP) antifouling blocks and terminal poly(dopamine-acrylamide) (PDAA) anchoring blocks were synthesized. These polymers can form a robust antifouling nanolayer on various surfaces. The morphologies of the polymer-modified surfaces are strongly dependent on the topologies of the polymers: with the increase of arm numbers, the morphology evolves from the smooth surface to the nanoscale coarse surface. As a result, the hydrophilicity of the coatings increases with the increase of degree of nanoscale roughness, and the 4-armed block copolymer forms a superhydrophilic surface with a water contact angle (WCA) as low as 8.7°. Accordingly, the linear diblock copolymer exhibits the worst antifouling efficiency, while the 4-armed polymer exhibits the best antifouling efficiency. This is the first example systematically showing that the antifouling efficacy could be adjusted simply by the topology of the coatings. Cell viability studies revealed that all of the copolymers exhibit excellent cytocompatibility. These biocompatible polymers with narrowly distributed molecular weight might find niches for antifouling applications in various areas such as anti-protein absorption, anti-bacterial and anti-marine fouling.

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Introduction

Polymers with antifouling properties have been used to modify surfaces of various devices, including but not limited to medical implants, biosensors, food packaging and marine equipment, to achieve desired antimicrobial/antibacterial efficacy.^{1–3} Poly(ethylene glycol) (PEG) has been studied extensively in the development of antifouling coatings due to its hydrophilicity and nontoxicity.^{4,5} However, as a polyether, PEG may undergo oxidative degradation and lead to chain scission when placed in complex biological fluids or in salt water media, which is the major limitation for its long-term applications.⁶ Therefore, antifouling polymers other than PEG which are stable toward oxidation and suitable for surface modification are important alternatives deserving further investigation.^{7–9}

Poly(*N*-vinylpyrrolidone) (PVP) is a widely used and readily available biocompatible polymer also known for its antifouling properties.⁷ Compared with PEG, PVP is more stable against

oxidizers and more preferred for long-term use.^{6,10} For example, PVP materials can resist oxidation to maintain their native structure even in 3% of H₂O₂ solution for 100 h.^{10–12} Notably, Serrano *et al.* have demonstrated that when grafted onto a surface by reacting with *in situ* generated nitrene, PVP exhibits more consistent resistance against all bio-assays than PEG, which might be suitable for marine non-fouling applications.¹³ Telford *et al.* reported that PVP-modified surfaces exhibit lower fibrinogen adsorption than PEG-modified surfaces.¹⁴ Nevertheless, the modification methods of PVP to surfaces/membranes are limited to physical absorption, blending, UV or γ -ray radiated grafting and surface initiated free-radical polymerization.^{15–20} However, these methods have their own drawbacks: physical absorption of PVP onto surfaces suffers from weak adhesion; blending PVP to other polymers suffers from leaking; UV or γ -ray radiated grafting of PVP might not be applicable for objects with complex inner surfaces, and surface initiated free-radical polymerization is complicated and suffers from low grafting rate, which limits their practical applications. So, the development of simple, universal and robust modification methods for antifouling surfaces based on PVP is still challenging but highly desirable.

Antifouling PVP coatings suitable for various surfaces require strong anchoring points that could withstand high load and shear stress under marine or physiological conditions. Inspired by the remarkable underwater adhesion capability of mussels, much effort has been dedicated to the preparation of

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antifouling polymers conjugated to mussel-mimetic polymeric anchors.²¹ The strong H-bonding formation and complexation capabilities of catechol moieties with various surfaces help these polymers to form robust antifouling layers.^{22,23} Many antifouling ingredients/groups such as oligo(ethylene glycol) (OEG),^{24,25} zwitterions,^{26,27} peptides,²⁸ perfluorinated alkyl chains,²⁹ and silver nanoparticles³⁰ have been integrated with catechol moieties. However, only a few reports focused on PVP antifouling polymers. For example, Lee *et al.* synthesized catechol-conjugated PVP by quarternizing the amide group on the pyrrolidone structure using 2-chloro-3,4'-hydroxyacetophenone, which can be securely and effectively coated on glass (GS), gold, and tissue culture polystyrene surfaces.³¹ The coated surfaces became more hydrophilic and protein-resistant due to the presence of PVP. Recently, this group synthesized a catechol-functionalized comb-shaped copolymer *via* grafting carboxylic-terminated PVP and caffeic acid onto the ϵ -polylysine backbone, which could be easily coated on polypropylene (PP) and poly-(dimethylsiloxane) membranes and exhibits excellent antifouling performance.^{32,33} However, these reports generally treat PVP as an additional antifouling ingredient, without considering the effect of the topology of PVP segments on antifouling performance.

In fact, the aggregation behavior of hydrophilic polymers greatly affects the antifouling performance of the coating. Lubricin, a natural ABA-type triblock protein, is known to adopt a loop conformation on surfaces, displaying excellent lubrication and antifouling properties.^{34,35} Quite a few studies have also confirmed that when PEG, poly(2-ethyl-2-oxazoline) or zwitterionic polymer is used as a hydrophilic antifouling segment, the polymers that adopt loop or cyclic-conformation on the surface possess better antifouling properties compared to those forming traditional brush-like conformation.^{36–42} The superior antifouling properties are mainly attributed to the strong steric hindrance of the loop and cyclic conformations and the lubricating behavior owing to the absence of the chain end. However, the influence of more sophisticated topologies on antifouling properties, to the best of our knowledge, has not been studied. Inspired by these reports, we decided to explore further the influence of the topologies of the *N*-vinylpyrrolidone (VP) block copolymers on the outcome of the antifouling properties. Herein, we present for the first time the synthesis of 3-armed and 4-armed polymers (each arm consists of a diblock structure containing a PVP block and a terminal poly(dopamine-acrylamide) (PDAA) anchoring block) *via* reversible addition-fragmentation chain transfer (RAFT) polymerization and compared their antifouling effect with traditional diblock (which forms brush on surfaces) and triblock copolymers (which forms loop on surfaces) containing PVP and PDAA. Well-defined polymers with narrow molecular weight distribution were obtained. We envision that for 3- and 4-armed block copolymers, three to four anchoring points to the surface could be formed, and the polymer would adopt a more complicated conformation on the surface than those linear and loop polymers, which would eventually influence their antifouling properties. We found that the morphologies of these coatings are obviously controlled by the topologies of the block copolymers: the linear diblock copolymer forms a nanoscale smooth surface, which is

hydrophilic, while the 4-armed block copolymer forms a nanoscale coarse surface, which is superhydrophilic. As a result, the antifouling properties of the polymer-modified surfaces are strongly dependent on the topologies of the polymers, in which the diblock copolymer exhibits the worst antifouling efficiency, while the 4-armed polymer exhibits the best antifouling efficiency. To the best of our knowledge, this is the first example systematically showing that the antifouling efficacy could be adjusted simply by the topology of the coatings. These biocompatible polymers with narrowly distributed molecular weight might find niches for antifouling applications in various areas such as anti-protein absorption, anti-bacterial and anti-marine fouling.

Experimental section

Materials

Ethyl 2-bromopropionate (EBP), potassium ethyl xanthogenate, 2-bromopropionyl bromide, acryloyl chloride and dopamine hydrochloride were purchased from Sigma and used as received. VP and ethylene glycol (EG) were distilled and stored using molecular sieves before use. Pentaerythritol (PE), trimethylolpropane (TMP) and 2,2'-azo-bis(2-methylpropionitrile) (AIBN) were recrystallized from water, ethyl acetate and ethanol, respectively. Albumin-fluorescein isothiocyanate conjugate and bovine serum albumin (BSA-FITC), Luria-Bertani (LB) broth and LB agar were purchased from Sigma. *Staphylococcus aureus* (*S. aureus*, ATCC 25923) and *Escherichia coli* (*E. coli*, ATCC 25922) were obtained from China Pharmaceutical Culture Collection. Dopamine-acrylamide (DAA) was synthesized according to the previously published work.⁴³

Synthesis of block copolymers with diverse topologies

Mussel-inspired catechol-functionalized PDAA and PVP were selected as the anchoring A block and antifouling B block, respectively. Four kinds of chain transfer agents named EBP-CTA, EG-CTA, TMP-CTA and PE-CTA were synthesized from potassium ethyl xanthogenate and the corresponding bromine reagent (see Scheme S1 and Fig. S1–S7 in the ESI,[†] for the detailed synthesis process and characterization). Block copolymers with different topologies (A-b-B, A-b-B-b-A), and S(A-b-B) (star shaped A-b-B) were prepared by RAFT polymerization, using EBP-CTA, EG-CTA, TMP-CTA and PE-CTA as chain transfer agents, respectively.

Synthesis of macro-chain transfer agent. Twelve macro-chain transfer agents, namely PVP_{*m*}-CTA, CTA-PVP_{*2m*}-CTA, 3-S(PVP_{*m*}-CTA) and 4-S(PVP_{*m*}-CTA) with three different molecular weights of PVP blocks in each category were synthesized. A typical synthetic procedure, using PVP₅₁-CTA as an example, is shown below: AIBN (8.2 mg, 0.05 mmol), EBP-CTA (55.6 mg, 0.25 mmol), VP (1.40 g, 12.5 mmol) and dioxane (6.0 mL) were charged into a 25 mL dry Schlenk tube equipped with a magnetic bar under a nitrogen atmosphere. The tube was connected to a standard Schlenk line, and the reaction mixture was degassed *via* three freeze–pump–thaw cycles and then immersed in an oil bath at 60 °C for 24 h. The polymerization was stopped by freezing the

reaction mixture with liquid N₂. The mixture was precipitated into diethyl ether, and the precipitate was collected on a filter and dried under reduced pressure, and 1.16 g PVP₅₁-CTA was obtained (yield: 80%). By adjusting the chain transfer agent and the feed ratio, other macro-chain transfer agents with different molecular weights of PVP were obtained in yield more than 80%. The ¹H NMR spectra are shown in Fig. S9 and S20 (ESI†).

Synthesis of block copolymers. Block copolymers with different topologies were synthesized by the RAFT polymerization of DAA using PVP-CTA (or CTA-PVP-CTA, or 3-S(PVP-CTA) and or 4-S(PVP-CTA)) as the macro-chain transfer agent. Four categories of copolymers, namely PVP_{*m*}-PDAA₁₀, PDAA₁₀-PVP_{2*m*}-PDAA₁₀, 3-armed (PVP_{*m*}-PDAA₁₀), and 4-armed (PVP_{*m*}-PDAA₁₀), with three different molecular weights of PVP blocks in each category and fixed repeat units of DAA were synthesized (totally 12 polymers). A typical synthetic procedure, using the synthesis of PVP₅₁-PDAA₁₀ as an example, is shown below: PVP₅₁-CTA macro-chain transfer agent (1.76 g, 0.3 mmol), AIBN (16.4 mg, 0.1 mmol) and DAA (0.74 g, 3.6 mmol) were dissolved in DMF (8 mL) at room temperature and the mixture solution was degassed *via* nitrogen purge for 30 min and then stirred vigorously under nitrogen at 60 °C for 24 h. The mixture was precipitated into diethyl ether and washed sequentially with methanol and then dried under reduced pressure, and PVP₅₁-PDAA₁₀ was obtained in 90% yield (2.25 g). By adjusting the macro-chain transfer agent and the feed ratio, other block copolymers with diverse topologies were obtained in yield more than 90%. The ¹H NMR spectra are shown in Fig. S21–S32 (ESI†).

Cytotoxicity of block copolymers with diverse topologies

The biocompatibility of block copolymers with diverse topologies against Hep G2 cells was evaluated *in vitro* by MTT assays. Hep G2 cells were seeded in a 96-well plate with a density of 10 000 cells per well and incubated in DMEM (100 µL) for 24 h. The medium was then replaced with 100 µL of serial dilutions of block copolymer solutions in DMEM (containing <0.1% DMSO for the dissolution of the copolymers). The cells grew for another 24 h. Then, 10 µL of MTT solution in PBS with the concentration of 5 mg mL^{−1} was added and the plate was incubated for another 4 h at 37 °C. After that, the medium containing MTT was removed and 150 µL of DMSO was added to each well to dissolve the MTT formazan crystals. Finally, the plate was shaken for 10 min, and the absorbance of the formazan product was measured at 492 nm using a Perlong-DNM-9602 microplate reader.

Deposition of block copolymer coatings on the GS substrate

The GS substrate was used for antifouling film preparation. The GS was cut into pieces with 1 cm × 1 cm size and cleaned ultrasonically with doubly distilled water, acetone and ethanol for 15 min each. The substrates were then rinsed thoroughly with doubly distilled water and blown dry with compressed air. Block copolymers with different topologies were dissolved in methanol with the concentration of 10 mg mL^{−1}. Each polymer solution was added dropwise onto the GS surface. After 30 minutes of adsorption, the surface was rinsed with methanol to remove unbound

molecules and dried in a vacuum. The resulting PVP_{*m*}-PDAA₁₀, PDAA₁₀-PVP_{2*m*}-PDAA₁₀, 3-armed (PVP_{*m*}-PDAA₁₀), and 4-armed (PVP_{*m*}-PDAA₁₀) anchored substrates were denoted as GS-PVP_{*m*}-PDAA₁₀, GS-PDAA₁₀-PVP_{2*m*}-PDAA₁₀, GS-3-armed (PVP_{*m*}-PDAA₁₀) and GS-4-armed (PVP_{*m*}-PDAA₁₀), respectively. The final coatings were then used for the following test. Uncoated glass disks were used as controls. Other substrates (stainless steel (SS), PP and mica) were also used to evaluate the binding ability of block copolymers.

Protein adsorption on the block copolymer-anchored glass substrates

BSA-FITC was dissolved in phosphate buffered saline (PBS, 10 mM, pH 7.4) to a protein concentration of 2 mg mL^{−1}. The block copolymer-anchored substrates were placed in a 24-well plate and immersed with the dissolved protein solution for 12 h at 37 °C. The plate was capped and wrapped with an aluminum foil to provide a dark environment. The substrates were then washed twice with PBS and once with doubly distilled water to remove weakly adhered BSA-FITC. The substrates were then viewed under a Nikon Eclipse Ti microscope equipped with a 470 nm excitation filter and a 540 nm emission filter for green fluorescence. For quantitative analysis, NIH Image J software was used to analyze the images obtained *via* fluorescence microscopy. The fluorescence intensity, which is indicative of the amount of protein adsorbed, was quantified on a per-pixel basis, summed up, and divided by the total number of pixels on the image to obtain mean values.

Bacterial adhesion to the block copolymer-anchored glass substrates

S. aureus and *E. coli* were incubated on separated agar plates (LB nutrient medium containing 1.5 wt% agars) overnight at 37 °C. A single colony of each bacterium was used to inoculate 25 mL of LB medium, and cultured overnight at 37 °C by shaking. For the bacterial adhesion assay, the bacteria containing growth broth was collected by centrifugation and subsequently suspended in PBS to get a concentration of 10⁸ cells mL^{−1} (the optical density (O. D.) reading of 0.1 at the wavelength of 540 nm on a micro-plate reader corresponds to the concentration of ~10⁸ cells per mL). The samples (size: 1.0 cm × 1.0 cm) were placed in a 24-well plate and incubated with 1 mL of bacterial suspension at 37 °C for 4 h. Quantification of bacteria adhesion on the pristine and grafted substrates was determined using the spread plate method. The bacteria suspension in each well was gently removed by aspiration using a pipette, and 1 mL of deionized water was slowly added to each well along the wall. The deionized water was then gently pipetted away. After washing with deionized water three times, the substrates were put into 3.0 mL of deionized water and subjected to ultrasonication for 7 min, followed by vortexing for 20 s to release the cells. The bacterial solution was then serially diluted, spread on an agar plate, and cultured overnight to quantify the number of bacterial cells. All experiments were performed in triplicate with three samples and the mean values were calculated. The number of adherent

bacterial cells was expressed as cells per mm^2 of the substrate surface.

Characterization

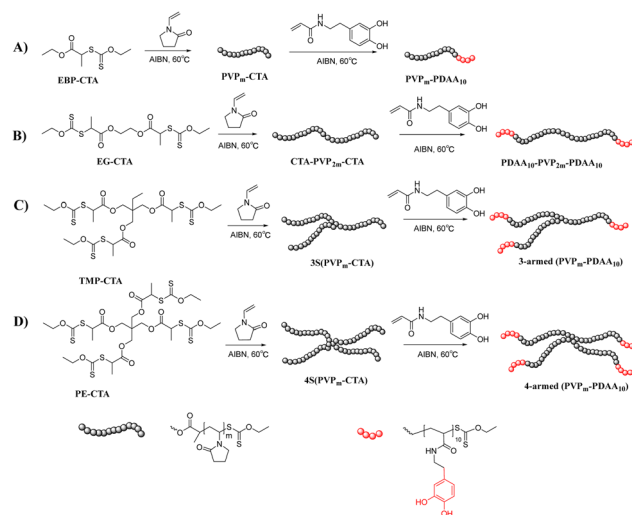
The chemical structures of the obtained monomers and polymers were characterized by ^1H NMR spectroscopy and the ^1H NMR spectra were recorded on a Bruker AV400 MHz spectrometer using TMS as an internal standard. Molecular weight and polydispersity index (PDI) of the polymers were determined on a gel permeation chromatograph (GPC, PL-GPC 50 Plus, Varian) with DMSO containing 0.01 M LiBr as the eluent (flow rate: 0.5 mL min^{-1} , at 50°C) against poly(methyl methacrylate) standards. Morphologies of coating surfaces were observed *via* AFM. AFM measurements were performed on an Agilent 5400 instrument using the tapping mode. The average roughness of pristine and modified coating surfaces was determined from AFM topographies by averaging over five random locations on each surface. The thickness of the coating was measured by a step profiler (Veeco Dektak 150) with a scan length of $100\text{ }\mu\text{m}$ at a scan speed of $10\text{ }\mu\text{m s}^{-1}$, after scraping off few parts of the coating film using a razor blade. Five different areas were measured for every sample surface and the average value of the thickness was adopted. Water contact angles of film surfaces were measured at ambient temperature on a Contact Angle Meter (DSA100, Kruss Company, Germany). Water droplets of $5\text{ }\mu\text{L}$ were dropped carefully on the coating surfaces. Each of the reported contact angles represents the average value of the five measurements.

Results and discussion

Synthesis and characterization of block copolymers with different topologies by using RAFT polymerization

Previously, we reported the random copolymerization of VP with a catechol-bearing VP monomer, 3,4-bis-hydroxyl-benzyl-*N*-vinyl-2-pyrrolidone, and the polymer exhibits strong underwater adhesion capability.^{44,45} However, the random structure greatly decreases the hydrophilicity of the copolymer and makes it unsuitable for antifouling application. We envisioned that the block copolymer of VP with a catechol-containing monomer would exhibit both strong adhesion to various surfaces and antifouling properties. RAFT polymerization techniques are powerful tools to synthesize block copolymers with a predetermined architecture, molecular weight and a narrow dispersion.⁴⁶ In RAFT polymerization, the choice of the chain transfer agent (CTA) is based on the monomers and is crucial for controlling the polymerization kinetics and the molecular weight of the polymer. VP is a representative of non-conjugated monomers and its radicals are poorly stable. Therefore, the Z group of the CTA must also be poorly stabilized in order to enable fast fragmentation of the polymer growing chains. This is typically done using a xanthate or dithiocarbamate CTA due to their reduced reactivity, allowing both faster radical addition and fragmentation.

In this study, four kinds of xanthate CTAs with different topologies were first obtained *via* the nucleophilic substitution



Scheme 1 The RAFT polymerization of block copolymers with different topologies, (A) $\text{PVP}_m\text{-PDAA}_{10}$, (B) $\text{PDAA}_{10}\text{-PVP}_{2m}\text{-PDAA}_{10}$, (C) 3-armed ($\text{PVP}_m\text{-PDAA}_{10}$) and (D) 4-armed ($\text{PVP}_m\text{-PDAA}_{10}$)

reaction between potassium *O*-ethyl xanthate and the corresponding 2-bromo-propanoic esters (Scheme S1, ESI†). The corresponding mono-functional, di-functional, 3-armed and 4-armed xanthate CTAs (denoted as EBP-CTA, EG-CTA, TMP-CTA and PE-CTA, respectively) were characterized by ^1H NMR (Fig. S1–S7, ESI†). Then, a series of block copolymers with fixed DAA repeat units (10 units for each end) but different topologies and VP lengths were prepared accordingly (Scheme 1 and Table 1). The block copolymers were named as $\text{PVP}_m\text{-PDAA}_{10}$, $\text{PDAA}_{10}\text{-PVP}_{2m}\text{-PDAA}_{10}$, 3-armed ($\text{PVP}_m\text{-PDAA}_{10}$), and 4-armed ($\text{PVP}_m\text{-PDAA}_{10}$), where *m* and 10 represent the repeating units' number of each block determined by ^1H NMR.

In order to obtain well-defined block copolymers, the kinetics of polymerization of VP with different CTAs (EBP-CTA, EG-CTA, TMP-CTA and PE-CTA) was evaluated by monitoring the conversion rate by the ^1H NMR study at each reaction time point on a

Table 1 Characterization of block copolymers of different topologies

Run	Composition ^a	M_n (KDa)		D^d
		GPC ^b	NMR ^c	
1	$\text{PVP}_{51}\text{-PDAA}_{10}$	12	7.7	1.10
2	$\text{PVP}_{116}\text{-PDAA}_{10}$	22	15	1.15
3	$\text{PVP}_{212}\text{-PDAA}_{10}$	33	26	1.15
4	$\text{PDAA}_{10}\text{-PVP}_{128}\text{-PDAA}_{10}$	23	18	1.20
5	$\text{PDAA}_{10}\text{-PVP}_{218}\text{-PDAA}_{10}$	41	28	1.21
6	$\text{PDAA}_{10}\text{-PVP}_{404}\text{-PDAA}_{10}$	59	49	1.19
7	3-armed ($\text{PVP}_{52}\text{-PDAA}_{10}$)	27	24	1.21
8	3-armed ($\text{PVP}_{104}\text{-PDAA}_{10}$)	59	41	1.19
9	3-armed ($\text{PVP}_{202}\text{-PDAA}_{10}$)	99	74	1.20
10	4-armed ($\text{PVP}_{53}\text{-PDAA}_{10}$)	32	32	1.16
11	4-armed ($\text{PVP}_{110}\text{-PDAA}_{10}$)	66	58	1.22
12	4-armed ($\text{PVP}_{204}\text{-PDAA}_{10}$)	106	100	1.20

^a The subscript numbers represent the repeating units' number of each block determined by ^1H NMR. ^b M_n determined by gel permeation chromatography (GPC) using poly(methyl methacrylate) standards for calibration. ^c M_n calculated from ^1H NMR measurements. ^d D represents the molecular weight distribution index.

given reaction time scale. The measurements were based on the integration values of vinyl protons (7.08 ppm, 1H for NVP) of unreacted monomers and the pyrrolidone methylene protons (1.0–2.6 ppm, 6H per unit for PVP plus 4H for VP) (Fig. S8, ESI†). Fig. 1A shows a plot of the semi-logarithmic $\ln([M]_0/[M]_t)$ versus the polymerization time of VP. A linear dependence was maintained up to a very high monomer conversion at 60 °C when EBP-CTA was used as a monofunctional CTA. The pseudo-first-order kinetics of the RAFT polymerization process suggested that active radical concentrations were kept constant throughout the polymerization process. When multifunctional chain transfer agents (EG-CTA, TMP-CTA and PE-CTA) were used, the polymerization rate increased with the increased functionality of the chain transfer agent at the same VP original concentration and targeted DP (degree of polymerization), and the kinetic plot showed an obvious deviation from the linear relationship of pseudo-first-order kinetics at the later stage of polymerization, especially for the star-shaped PVP-CTA. It is obvious that when the VP original concentration and targeted DP remain unchanged, the molar content of the xanthate group increases in the order of EBP-CTA < EG-CTA < TMP-CTA < PE-CTA. Hence, the polymerization rate increases with the increased functionality of the chain transfer agent. The deviation of the kinetic plot for CTA-PVP-CTA and star-shaped PVP-CTA could be explained by the increased solution viscosity and the decreased activity of the propagating radicals at the later stage of polymerization, which lowers the polymerization rate and results in negative deviation from linearity of pseudo-first-order kinetics.

A series of PVP macro-chain transfer agents named PVP_m-CTA, CTA-PVP_{2m}-CTA, 3-S(PVP_m-CTA) and 4-S(PVP_m-CTA) (where *m* represents the repeating units' number of PVP determined by ¹H NMR) with different topologies and molecule weights were easily prepared. The ¹H NMR technique was used to characterize the number average molecular weight (*M_n*) of different PVP macro-chain transfer agents. A typical ¹H NMR spectrum of PVP_m-CTA is shown in Fig. 1B and all the ¹H NMR spectra of macro-chain transfer agents are shown in Fig. S9 and S20 (ESI†). DP was estimated using the ratio of peak integration of the proton in PVP_m (δ 3.0–4.1 ppm, 3 mH) to the peak

integration of the methylene proton in the xanthate ending group (δ 4.60 ppm, 2H). The results are listed in Table S1 (ESI†). GPC was used to determine the molecular weight distribution (Table S1, ESI†) and the typical GPC curves of each of macro-chain transfer agents are shown in Fig. 1C. As shown, the GPC curves showed symmetrical and unimodal elution peaks and no shoulder peaks or tailings were observed, denoting well-defined PVP macro-chain transfer agents.

With the obtained xanthate PVP macro-chain transfer agents in hand, the next step is to use it to initiate the polymerization of a catechol-containing monomer. Initially, a VP monomer bearing a catechol moiety, *i.e.*, 3,4-bis-hydroxyl-benzyl-*N*-vinyl-2-pyrrolidone, was selected for the block polymerization. Unfortunately, although different conditions (including adjusting the polymerization temperature, [CTA]₀/[Initiator]₀ molar ratio and using the monomer with the protected phenyl group) were tried, no initiation of the block polymerization was observed. We speculated that the attachment of the catechol moiety to VP may greatly increase the steric hindrance and decreases its reactivity toward a macro-chain transfer reagent, which retards the block polymerization. Hence, acrylamide derivatives, which show a similar polymerization capability to VP, were then selected for the block polymerization.

DAA turned out to be a successful chain propagation monomer for all PVP macro-chain transfer agents with different topologies. The molecular weights and molecular weight distribution of the synthesized block copolymers were characterized by ¹H NMR spectra (Fig. 1D and Fig. S21–S32, ESI†) and GPC (Table 1 and Fig. 1C), respectively. As shown in Fig. 1D, new characteristic proton signals related to PDAA blocks appeared at 6.2–6.8 and 8.4–8.8 ppm, which were assigned to phenyl protons and hydroxyl protons of PDAA blocks, respectively. The integral ratio of the proton signals on PDAA repeating phenyl protons (δ 6.2–6.8 ppm) to the methylene protons in the xanthate-end group (δ 4.18 ppm) was used to calculate the number of the DAA repeat units on the block copolymers, and the results are listed in Table 1. All the calculated DAA repeat units showed similarity in terms of the feeding ratio of DAA to PVP macro-chain transfer agents, indicating that the reactivity of DAA is matched with that of the PVP macro-chain transfer reagent. GPC was used to determine the molecular weight distribution (Fig. 1C). The GPC curves showed almost symmetrical and unimodal elution peaks and no obvious shoulder peaks or tailings were observed, denoting well-defined block polymers. Taking PVP₁₁₆-PDAA₁₀ as an example (Fig. 1C insert), the retention time was about 14.4 min, which was slightly shorter than that of PVP₁₁₆-CTA (15.3 min). The *M_n* values of PVP₁₁₆-PDAA₁₀ and PVP₁₁₆-CTA calculated *via* GPC were 22 kDa and 15 kDa, respectively. These results indicated the successful chain extension of PVP₁₁₆-CTA by using the DAA monomer. It is noteworthy to mention that the symmetry of the GPC curve of PVP₁₁₆-PDAA₁₀ was not as good as that of PVP₁₁₆-CTA and the *Đ* was a little bit higher. These results may be attributed to the polymerization sequence that was not the optimal one due to the topology and functionality was the first-considered.

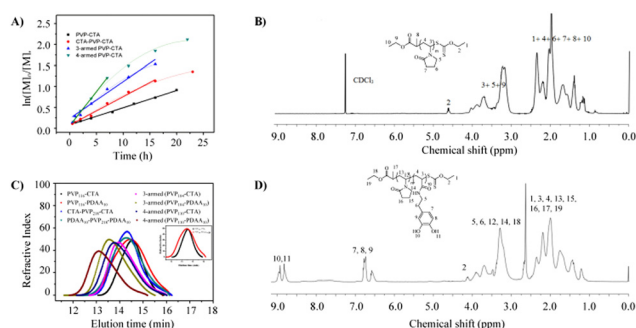


Fig. 1 (A) Kinetic studies on the polymerization of NVP with different chain transfer agents, (B) the ¹H NMR spectrum of PVP_m-CTA in CDCl₃, (C) the typical GPC curves of each of macro-chain transfer agent and the corresponding block copolymers, and (D) the ¹H NMR spectrum of PVP_m-PDAA₁₀ in DMSO-*d*₆.

Surface characterization

Using a simple solution dipping method, the block copolymer could be adsorbed onto the substrate due to the existence of the terminal catechol groups. Block copolymers with different topologies were dissolved in methanol with the concentration of 10 mg mL^{-1} for surface modification. It should be noted that PVP/PDAA ratios were kept constant in each assay so that the influence of topologies could be compared. For example, in an assay containing GS-PVP₁₁₆-PDAA₁₀, GS-PDAA₁₀-PVP₂₁₈-PDAA₁₀, GS-3-armed (PVP₁₀₄-PDAA₁₀) and GS-4-armed (PVP₁₁₀-PDAA₁₀), the PVP/PDAA ratio is close to 10:1, so that in a 10 mg mL^{-1} solution containing the polymer with linear topology, the overall amount of PDAA and PVP is almost identical with the solution containing the polymer with 4-arm topology.

For the block copolymers with different topologies, we anticipated that brush, loop, tripod and tetrapod topologies can be formed on the surface with the increase of arm numbers (Fig. 2A). The morphology and the nanostructure of the block copolymer-anchored surfaces were studied *via* atomic force microscopy (AFM). All of the block copolymers were uniformly distributed on the surface of GS, revealing the critical role of the free catechol moieties in the anchoring of the copolymers on the substrate (Fig. 2B and Fig. S33, ESI†). The brush-like and loop-like surfaces formed by PVP_m-PDAA₁₀ and PDAA₁₀-PVP_{2m}-PDAA₁₀ are relatively smooth and flat with an average roughness of less than 1.0 nm (Table S2, ESI†), which is in agreement with other reported brush-bearing and loop-bearing surfaces prepared by catechol-functionalized PEG-based triblock/diblock copolymers.^{38,40–42} There is no obvious difference in morphology between the polymers with the same topology but with different length of PVP segments (Fig. S33 and Table S2, ESI†). Quite surprisingly, distinctive morphologies can be observed on GS-3-armed (PVP_m-PDAA₁₀) and GS-4-armed (PVP_m-PDAA₁₀) surfaces

(Fig. 2B and Fig. S33, ESI†). The average roughness of these surfaces (1–2 nm for 3-armed (PVP_m-PDAA₁₀) anchored surfaces and 3–4 nm for 4-armed (PVP_m-PDAA₁₀) anchored surfaces) is obviously higher than that of GS-PVP_m-PDAA₁₀ and GS-PDAA₁₀-PVP_{2m}-PDAA₁₀ surfaces (Table S2, ESI†). Likewise, there is no obvious morphological difference between the polymers with the same topology but with different length of PVP segments. All of the 3-armed polymers show a continuous mastoid surface, indicating that the arrangement of the molecular chains on the surface is vastly different compared with PVP_m-PDAA₁₀ and PDAA₁₀-PVP_{2m}-PDAA₁₀. All of the 4-armed (PVP_m-PDAA₁₀) anchored surfaces exhibit a continuous and regular wavy nanostructure, which is similar to the microstructure of the leaves of *Calathea zebrina* and *Ruellia devosiana*, a surface with a homogeneously conical micro-nanostructure, showing superhydrophilic and excellent antifouling ability.^{47,48} We can conclude that the morphologies of the coatings are majorly governed by the topologies, not by the molecular weight of the polymers. These coarse and special morphologies are markedly different from the other reported loop-bearing surfaces that exhibit surfaces smoother than brush-bearing surfaces due to the absence of chain ends.^{38,40–42} Such innovative morphologies are mainly due to the fact that 3-armed (PVP_m-PDAA₁₀) and 4-armed (PVP_m-PDAA₁₀) have multiple anchors, probably forming a tetrapod-like and tripod-like microstructure, respectively, which also contributes to surface hydrophilic and antifouling properties. The complex interaction between multiple anchors may be the main reason for the formation of these special morphologies. The thickness of the loop, brush, tripod and tetrapod layers was determined by the scratch test. With the similar length for each arm, thicker layers are formed for polymers with more arms. For instance, the highest PVP molecular weight for block copolymers (PVP₂₁₂-PDAA₁₀, PDAA₁₀-PVP₄₀₄-PDAA₁₀, 3-armed (PVP₂₀₂-PDAA₁₀) and 4-armed (PVP₂₀₄-PDAA₁₀)) was determined to be $4.52 \pm 0.36 \text{ nm}$, $8.54 \pm 0.41 \text{ nm}$, $14.62 \pm 0.64 \text{ nm}$, and $21.25 \pm 0.75 \text{ nm}$, respectively. We postulate that the smaller hydrodynamic radius of tripod- and tetrapod-like macromolecules enables the fabrication of denser surface grafting, leading to thicker coatings. For each morphology with different molecular weights of PVP, the greater the molecular weight, the thicker the thickness of the block copolymer-coated surface (Table S2, ESI†). For example, when the molecular weight of PDAA₁₀-PVP_{2m}-PDAA₁₀ increases from 28 kDa to 49 kDa, the coating thickness increases from 4.65 nm to 8.54 nm.

The differences of morphologies also result in different surface hydrophilicities, which were confirmed by water contact angle (WCA) measurements. Fig. 2C shows the static WCAs and the corresponding images of the block copolymer-anchored GS and pristine GS surfaces. All modified surfaces exhibit lower contact angles than that of the pristine GS surface due to the excellent hydrophilicity of PVP. More interestingly, the contact angles of block copolymer-anchored surfaces decrease as the copolymer topologies change from PVP_m-PDAA₁₀, PDAA₁₀-PVP_{2m}-PDAA₁₀, 3-armed (PVP_m-PDAA₁₀) to 4-armed (PVP_m-PDAA₁₀), which is mainly due to the difference in surface morphology. For example, the contact angles of pristine GS, GS-PVP₂₁₂-PDAA₁₀, GS-PDAA₁₀-PVP₄₀₄-PDAA₁₀, GS-3-armed

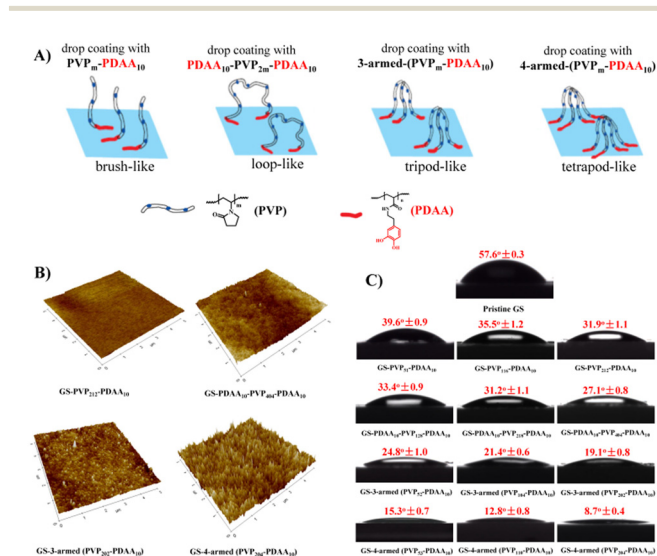


Fig. 2 (A) The microstructure of block copolymers with different topologies on the GS surface using the drop coating method, (B) AFM images of block copolymer-anchored GS surfaces, (C) the photomicrographs of the droplet attached on the pristine and modified GS surfaces, and the values over the droplet represent the average value of the WCA.

(PVP₂₀₂-PDAA₁₀) and GS-4-armed (PVP₂₀₄-PDAA₁₀) surfaces are 57.6°, 31.9°, 27.1°, 19.1°, and 8.7°, respectively. That is to say, the variation of block copolymer topologies can change the surface hydrophilicity of the anchored surface, in which the tetrapod-like surface prepared by 4-armed (PVP_m-PDAA₁₀) shows the best hydrophilicity, meeting the requirements of the superhydrophilic surface.⁴⁹ This significant change in WCA, especially for the GS-3-armed (PVP_m-PDAA₁₀) and GS-4-armed (PVP_m-PDAA₁₀) surfaces, is quite different from the reported loop-bearing surface. There was no obvious contact angle difference between brush-bearing and loop-bearing surfaces reported by Eeseul Shin³⁸ and Hanhan Wang.⁵⁰ Such a change in surface hydrophilicity is the synergic effect of nanoscale surface roughness and hydrophilicity of the polymer chains, which is a way to achieve superhydrophilicity.⁵¹ It should be noted that in many reported cases, the superhydrophilic surface is created by sophisticated surface-structure patterning such as laser irradiation or UV illumination using photoresponsive materials such as TiO₂. Other complicated techniques, such as layer-by-layer assembly, sol-gel assembly and etching can also be used.⁴⁹ In addition, our results provide a novel method wherein the superhydrophilicity of a coating can be simply controlled by the topologies of the coating polymers. To the best of our knowledge, such a micro-nanostructure adjustment for superhydrophilicity by polymer topology controlment has not been reported before. The difference in surface hydrophilicity can also lead to different antifouling properties. The surface with better hydrophilicity generally exhibits better anti-fouling performance against proteins or bacteria adhesion.^{52,53} It should also be noted that hydrophilicity is not only controlled by the surface nanostructure (morphology and topology) but also by the molecular weight of the hydrophilic segments in the polymers. For each of these topologies, the larger the molecular weight of PVP segments, the smaller the surface contact angle is. For example, with the increase of the length of PVP segments, the contact angle decreases from GS-PDAA₁₀-PVP₁₂₈-PDAA₁₀ (33.4°) to GS-PDAA₁₀-PVP₄₀₄-PDAA₁₀ (27.1°) in the loop structure, and the contact angle decreases from 4-armed (PVP₅₃-PDAA₁₀) (15.3°) to 4-armed (PVP₂₀₄-PDAA₁₀) (8.7°) in the tetrapod structure (4-armed block copolymers).

In addition to GS, other substrate materials, such as SS, PP and mica, were also chosen to verify the versatile surface binding ability of the block copolymer. The static contact angles of water droplets after coating different substrates with the block polymer provide a visual proof. Regardless of substrate hydrophilicity and hydrophobicity, the contact angle of the droplet on each coated surface was within a similar range of values, which demonstrated the potential of the catechol-functionalized block copolymer as a substrate-independent universal coating material (Fig. S34, ESI†). Moreover, the coatings always retain their original form even if they were rushed under the faucet for 10 min (Fig. S35, ESI†), indicating good adhesion on the substrate. Furthermore, the adhesion of the film to the substrate was evaluated using the cross-cut method, and the result showed that the adhesion is at the highest level according to the ASTM D3359 scale (Fig. 5B and Fig. S36, ESI†).

Protein adsorption on the block copolymer-anchored substrates

Nonspecific protein adsorption on surfaces, especially in medical implants and marine equipment applications, can reduce the efficacy of the devices and may also result in harmful side effects, such as immunological responses and the subsequent formation of a biofilm layer for the attachment of (marine) bacteria on the surfaces. Thus, surfaces that resist the nonspecific adsorption of protein and microbes are vital. Herein, bovine serum albumin (BSA) was used as a model to assess the protein resistance properties of the synthesized mussel-inspired block copolymers. The protein adsorption was examined by incubating the block copolymer-coated surfaces in a solution containing fluorescent BSA-FITC and subsequently imaged using a fluorescence microscope. Fig. 3 shows the respective fluorescence images and fluorescence intensities of the pristine GS, GS-PVP_m-PDAA₁₀, GS-PDAA₁₀-PVP_{2m}-PDAA₁₀, GS-3-armed (PVP_m-PDAA₁₀) and GS-4-armed (PVP_m-PDAA₁₀) surfaces after exposure to the PBS solution of BSA-FITC. Both the polymer chain length and the polymer topology were important factors for controlling the adsorption of proteins.

Feng *et al.* have reported that longer chains are needed to resist protein adsorption at a low graft density, while shorter chains with a high graft density are enough to resist proteins.⁵⁴ In our engineered surfaces, all the coatings exhibit uniform and dense surface morphology (Fig. 2B and Fig. S33, ESI†), implying that all surfaces have high graft density. Thus, the molecular weight of copolymers is a critical factor to be investigated for the anti-protein adsorption performance of the resulted coatings. As shown in Fig. 3, taking GS-PDAA₁₀-PVP₁₂₈-PDAA₁₀, GS-PDAA₁₀-PVP₂₁₈-PDAA₁₀ and GS-PDAA₁₀-PVP₄₀₄-PDAA₁₀ as an example, with the same topology, when the molecular weight of PDAA₁₀-PVP_{2m}-PDAA₁₀ increased from 18 kDa to 49 kDa, the amount of BSA-FITC adsorption decreased gradually, as the fluorescence intensities decreased from 4.95×10^5 to 4.07×10^5 . The reductions in the level of adsorption *versus* the unmodified GS surfaces ranged from 34% to 54%. Other engineered surfaces with same copolymer topologies show similar trends. These results showed that the protein adsorption on block copolymer-anchored surfaces was

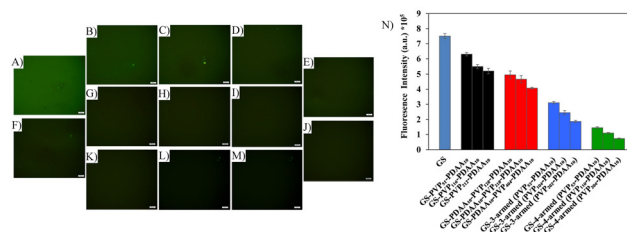


Fig. 3 Fluorescence microscopy images of the (A) pristine GS, (B) GS-PVP₅₁-PDAA₁₀, (C) GS-PVP₁₁₆-PDAA₁₀, (D) GS-PVP₂₁₂-PDAA₁₀, (E) GS-PDAA₁₀-PVP₁₂₈-PDAA₁₀, (F) GS-PDAA₁₀-PVP₂₁₈-PDAA₁₀, (G) GS-PDAA₁₀-PVP₄₀₄-PDAA₁₀, (H) GS-3-armed (PVP₅₂-PDAA₁₀), (I) GS-3-armed (PVP₁₀₄-PDAA₁₀), (J) GS-3-armed (PVP₂₀₂-PDAA₁₀), (K) GS-4-armed (PVP₅₃-PDAA₁₀), (L) GS-4-armed (PVP₁₁₀-PDAA₁₀) and (M) GS-4-armed (PVP₂₀₄-PDAA₁₀) surfaces after exposure to the BSA-FITC solution, (N) fluorescence intensities of the corresponding GS surfaces. Scale bar is 50 μ m.

strongly dependent on the molecular weight of block copolymers. The results were also in accordance with the literature report that the level of protein adsorption decreased significantly when the antifouling polymer chain length increased.⁵⁴

More importantly, the antifouling properties are strongly dependent on the topologies of the polymeric coatings. Both Li *et al.* and Kang *et al.*'s work showed that polymers with adhesive anchors on both chain ends (ABA type triblock copolymers, when contact with a surface, loops were formed) possessed better antifouling performance than those with only one end-tether (AB type diblock copolymers when contacted with a surface formed brushes).^{39,40} The superior antifouling properties of polymer loops may attribute to the strong steric hindrance compared with the brushes. On the basis of previous work, we were curious that when we further increased the steric hindrance (star-shaped copolymers), the antifouling properties would be or would not be enhanced. As shown in Fig. 3, the influence of topologies on the antifouling properties was obvious. Uniform and intense fluorescence is observed across the pristine GS surface (Fig. 3A) and the fluorescence intensity calculated as shown in Fig. 3A was 7.61×10^5 , suggesting a significant adsorption of BSA-FITC on the substrate surface. As the topologies of the grafted copolymers change from linear to star-shaped, the BSA-FITC adsorption decreases. In Fig. 3, taking GS-PVP₂₁₂-PDAA₁₀, GS-PDAA₁₀-PVP₄₀₄-PDAA₁₀, GS-3-armed (PVP₂₀₂-PDAA₁₀), and GS-4-armed (PVP₂₀₄-PDAA₁₀) as an example, the topology of the grafted copolymer is AB type, ABA type, 3-armed type and 4-armed type, respectively. As the topology changes, the amount of BSA-FITC adsorption decreased and the fluorescence intensities were 7.61×10^5 , 5.21×10^5 , 4.07×10^5 , 1.86×10^5 and 0.73×10^5 for pristine GS, GS-PVP₂₁₂-PDAA₁₀, GS-PDAA₁₀-PVP₄₀₄-PDAA₁₀, GS-3-armed (PVP₂₀₂-PDAA₁₀) and GS-4-armed (PVP₂₀₄-PDAA₁₀), respectively. The reductions in the level of adsorption *versus* the unmodified GS surfaces range from 31% to 91%. These results indicated that the adsorption of BSA-FITC protein on the block copolymer-anchored GS surfaces is significantly inhibited and the protein adsorption is strongly dependent on the grafted copolymer topologies. The unprecedented traits derived from the translation of topology effects by 3-armed and 4-armed copolymers into a marked alteration of relevant surface properties, namely (i) the smaller hydrodynamic radius of star-shaped macromolecules enables the fabrication of denser surface grafting and thicker coating (the average mass of the copolymers forming the coating is shown in Table S2, ESI†), (ii) their tripod-like and tetrapod-like topologies provide the same but stronger steric hindrance as loop-bearing surfaces, and (iii) better hydrophilicity derived from their coarse and special morphologies is also a crucial factor. To confirm whether the antifouling ability is mainly due to the grafting density or multi-arm structures, two series of coatings (one series has similar thickness but different topology, the other series has same topology but different thickness) were prepared by adjusting the concentration of the block copolymers (Table S3, ESI†). The anti-protein adsorption performances of the resulted coatings were tested and the results are shown in Fig. S37 (ESI†). As the topology changes, the amount of BSA-FITC adsorption on the

coating with the similar thickness showed a significant decreasing trend (Fig. S37A, ESI†). However, only a slight decrease could be found as shown in Fig. S37B (ESI†) for the coatings with same topology but different thicknesses. These results indicated that although the increase of graft density is beneficial to the antifouling ability of the coatings, the change of copolymer topologies is the main factor leading to the excellent antifouling ability of the coatings in our case. We concluded from our experiments that when we further increased the steric hindrance and hydrophilicity of the coating, the antifouling properties would be enhanced.

Bacterial adhesion on the block copolymer-anchored substrates

To further examine the antifouling properties of the engineered GS surfaces, the resistance to bacterial adhesion on the pristine GS, GS-PVP_m-PDAA₁₀, GS-PDAA₁₀-PVP_{2m}-PDAA₁₀, GS-3-armed (PVP_m-PDAA₁₀) and GS-4-armed (PVP_m-PDAA₁₀) surfaces was assayed. *E. coli* and *S. aureus* were chosen for this purpose. A number of adhered bacteria on the pristine and engineered GS surfaces were assayed quantitatively using the spread plate method, as shown in Fig. 4. The lower the number, the higher the antifouling ability is. Compared to the pristine GS surfaces, the presence of PVP_m-PDAA₁₀, PDAA₁₀-PVP_{2m}-PDAA₁₀, 3-armed (PVP_m-PDAA₁₀) and 4-armed (PVP_m-PDAA₁₀) can significantly improve the efficiency of the resultant coatings to resist bacterial adhesion, and the antifouling efficiency is strongly dependent on the coating components. Like the anti-protein adsorption performance of the resulted coatings, the molecular weight and topologies of block copolymers are also the critical factors for the anti-bacterial adhesion of surfaces. As shown in Fig. 4, taking GS-PDAA₁₀-PVP₁₁₆-PDAA₁₀, GS-PDAA₁₀-PVP₂₁₈-PDAA₁₀ and GS-PDAA₁₀-PVP₄₀₄-PDAA₁₀ as an example, with the increase of the molecular weight of the block copolymer, the number of viable *E. coli* cells on the corresponding surfaces decreases to 52%, 49% and 44%, respectively, as compared to that of the pristine GS surface. Other engineered surfaces with same copolymer topologies show similar trends. Similar results are also observed on conducting the *S. aureus* tests. This is due to the longer molecular chains and accompanying better surface hydrophilicity (Table 1 and Fig. 2C). These results indicate that the adhesion of bacteria on the engineering GS surfaces is significantly inhibited due to the repulsive forces arising from the formation of a hydration layer on the hydrophilic surface.

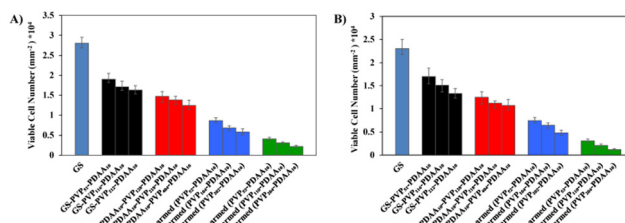


Fig. 4 Number of adhered (A) *E. coli* and (B) *S. aureus* cells per mm² of the pristine GS, GS-PVP_m-PDAA₁₀, GS-PDAA₁₀-PVP_{2m}-PDAA₁₀, GS-3-armed (PVP_m-PDAA₁₀) and GS-4-armed (PVP_m-PDAA₁₀) surfaces. Error bars give the standard deviation obtained from three replicates.

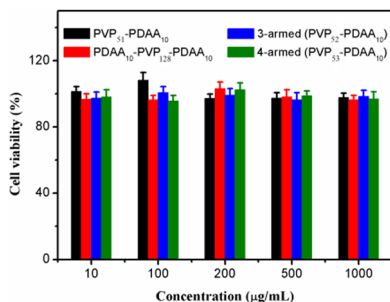


Fig. 5 Viability of Hep G2 cells after incubation with block copolymers at different dosages.

The topologies of block copolymers show greater influence for the anti-bacterial adhesion of surfaces, as compared to that of molecular weight. As shown in Fig. 4, take GS-PVP₁₁₆-PDAA₁₀, GS-PDAA₁₀-PVP₂₁₈-PDAA₁₀, GS-3-armed (PVP₁₀₄-PDAA₁₀), and GS-4-armed (PVP₁₁₀-PDAA₁₀) as an example, the number of viable *E. coli* cells on the corresponding surfaces decreases to 61%, 49%, 24% and 11%, respectively, compared to that of the pristine GS surface. Similar results are also observed on conducting the *S. aureus* tests. Regarding the geometrical conformation, the 4-armed copolymers present the strongest steric repulsion and the best antifouling effects compared to those of the corresponding diblock, triblock and 3-armed copolymers. Moreover, one can also find that the grafted coatings exhibit more effective antifouling activity for *S. aureus* than that for *E. coli*. This may be due to the complex cell membrane composition of *E. coli*.

In Vitro cytotoxicity of the block copolymers with different topologies

The cytotoxicity of the synthesized block copolymers (shown in Table 1) was evaluated against Hep G2 cells by the MTT assay. The viability of Hep G2 cells after incubation with block copolymers at different dosages from 10 µg mL⁻¹ to 1000 µg mL⁻¹ is shown in Fig. 5. For all the copolymers within the range of experimental concentrations showed excellent cytocompatibility. The outstanding biocompatibility enables the PDAA tethered PVP antifouling block copolymers to be widely used in medical implants, biosensors and marine equipment.

Conclusions

We have successfully presented for the first time the synthesis of diblock, triblock, 3-armed and 4-armed copolymers containing PVP blocks and terminal PDAA via RAFT polymerization, aiming at adjusting the topologies of these polymers when absorbed onto different surfaces. ¹H NMR and GPC were used to characterize the narrowly distributed finely-tuned polymers. The cytotoxicity of the synthesized block copolymers was evaluated against Hep G2 cells by MTT assays. Cell viability studies revealed that all the copolymers showed excellent cytocompatibility. By adjusting the PDAA/PVP ratio, a robust antifouling nano-layer could be formed on glass surfaces. We found that the antifouling properties of the polymer-modified surfaces are strongly dependent on the

topologies of the polymer, in which the diblock copolymer exhibits the worst antifouling efficiency, while the 4-armed polymer exhibits the best antifouling efficiency. These biocompatible polymers with narrowly distributed molecular weights might find niches for antifouling applications in various areas such as anti-protein absorption, antibacterial and anti-marine fouling. AFM and WCA studies revealed that with the evolution of the polymer topologies from a linear structure to a 4-armed structure, the nanoscale coarse surface could be obtained, which eventually leads to a superhydrophilic surface. Our results provide a novel method to construct superhydrophilic surfaces simply by controlling the topologies of the hydrophilic polymer. However, it is still unclear at this stage how the 3-armed and 4-armed polymers interact with the surface and form a morphology nanoscalely different from those linear and loop-type polymers. Deeper microscopic characterizations, such as the conformation study of the catechol functionalized copolymers with higher level topologies and their interactions with surfaces (especially at the single molecular level, both experimentally and theoretically), will cast light on the mechanism for the morphology control by topology, which will be our next work.

Author contributions

D. Pei, C. Xiong and W. Xiong performed the synthesis, tests and analysis experiments work. Y. Mu and X. Wan designed and supervised the research and wrote the manuscript.

Conflicts of interest

There are no conflicts to declare.

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